ASS 3

1. Link: https://github.com/Kur00o/CNL/blob/main/ASS3.1.c

```c
// ASS 3 - Subnetting


#include <stdio.h>


int main() {
   int a, b, c, d;


   printf("Enter IP Address: "); //192.168.10.5
   scanf("%d.%d.%d.%d", &a, &b, &c, &d);


   if (a < 0 || a > 255 ||
       b < 0 || b > 255 ||
       c < 0 || c > 255 ||
       d < 0 || d > 255) {
       printf("\nInvalid IP Address!\n");
       return 0;
   }


   if (a >= 0 && a <= 127) {
       printf("\nClass : A\n");
       printf("IP Range : 0.0.0.0 - 127.255.255.255\n\n");


       printf("Subnet Mask : 255.0.0.0\n");
```

```
    printf("A subnet mask is a 32-bit number used with an IP address to identify
    which part of the address represents the network and which part represents the
    host (device).\n\n");


    printf("\n32-bit Representation\n");

    printf("Network ID\n");

    printf("[0][N][N][N][N][N][N][N]");

    printf("[H][H][H][H][H][H][H][H]");

    printf("[H][H][H][H][H][H][H][H]");

    printf("[H][H][H][H][H][H][H][H]\n\n");


    printf("Hosts per Network ID : 2^24 - 2 = 16,777,214\n");

    printf("Calculation :\n");

    printf("Host bits = 24\n");

    printf("2^24 = 16,777,216\n");

    printf("Subtract 2 (Network ID and Broadcast Address)\n");

    printf("= 16,777,214 Hosts\n\n");


    printf("Number of Networks : 2^7 = 128\n");

    printf("Calculation :\n");

    printf("Network bits = 7\n");

    printf("2^7 = 128 Networks\n\n");


    if (a == 127)

        printf("Loopback Address : YES\n");

    else

        printf("Loopback Address : NO\n");
```

```
        printf("A loopback address is a special IP address that a computer uses to
        communicate with itself for testing and troubleshooting network software
        without sending data over a physical network.\n");

        printf("The purpose of a loopback address is to allow a computer to communicate
        with itself without using a physical network connection.\n");

    } else if (a >= 128 && a <= 191) {

        printf("\nClass : B\n");

        printf("IP Range : 128.0.0.0 - 191.255.255.255\n\n");


        printf("Default Subnet Mask : 255.255.0.0\n");

        printf("A subnet mask is a 32-bit number used with an IP address to identify
        which part of the address represents the network and which part represents the
        host (device).\n\n");

        printf("\n32-bit Representation\n");

        printf("[1][0][N][N][N][N][N][N]");

        printf("[N][N][N][N][N][N][N][N]");

        printf("[H][H][H][H][H][H][H][H]");

        printf("[H][H][H][H][H][H][H][H]\n\n");


        printf("Hosts per Network ID : 2^16 - 2 = 65,534\n");

        printf("Calculation :\n");

        printf("Host bits = 16\n");

        printf("2^16 = 65,536\n");

        printf("Subtract 2 (Network ID and Broadcast Address)\n");

        printf("= 65,534 Hosts\n\n");


        printf("Number of Networks : 2^14 = 16,384\n");

        printf("Calculation :\n");

        printf("Network bits = 14\n");

        printf("2^14 = 16,384 Networks\n\n");
```

```
    printf("Loopback Address : NO\n");

    printf("A loopback address is a special IP address that a computer uses to
    communicate with itself for testing and troubleshooting network software
    without sending data over a physical network.\n");

    printf("The purpose of a loopback address is to allow a computer to communicate
    with itself without using a physical network connection.\n");

} else if (a >= 192 && a <= 223) {

    printf("\nClass : C\n");

    printf("IP Range : 192.0.0.0 - 223.255.255.255\n\n");


    printf("Default Subnet Mask : 255.255.255.0\n");

    printf("A subnet mask is a 32-bit number used with an IP address to identify
    which part of the address represents the network and which part represents the
    host (device).\n\n");


    printf("\n32-bit Representation\n");

    printf("[1][1][0][N][N][N][N][N]");

    printf("[N][N][N][N][N][N][N][N]");

    printf("[N][N][N][N][N][N][N][N]");

    printf("[H][H][H][H][H][H][H][H]\n\n");


    printf("Hosts per Network ID : 2^8 - 2 = 254\n");

    printf("Calculation :\n");

    printf("Host bits = 8\n");

    printf("2^8 = 256\n");

    printf("Subtract 2 (Network ID and Broadcast Address)\n");

    printf("= 254 Hosts\n\n");


    printf("Number of Networks : 2^21 = 2,097,152\n");
```

```
        printf("Calculation :\n");

        printf("Network bits = 21\n");

        printf("2^21 = 2,097,152 Networks\n\n");


        printf("Loopback Address : NO\n");

        printf("A loopback address is a special IP address that a computer uses to
        communicate with itself for testing and troubleshooting network software
        without sending data over a physical network.\n");

        printf("The purpose of a loopback address is to allow a computer to communicate
        with itself without using a physical network connection.\n");

    } else if (a >= 224 && a <= 239) {

        printf("\nClass : D\n");

        printf("IP Range : 224.0.0.0 - 239.255.255.255\n\n");


        printf("Purpose : Multicast\n\n");


        printf("Default Subnet Mask : Not Applicable\n");

        printf("A subnet mask is a 32-bit number used with an IP address to identify
        which part of the address represents the network and which part represents the
        host (device).\n\n");


        printf("Hosts per Network ID : Not Applicable\n");

        printf("Number of Networks : Not Applicable\n\n");


        printf("Loopback Address : NO\n");

        printf("A loopback address is a special IP address that a computer uses to
        communicate with itself for testing and troubleshooting network software
        without sending data over a physical network.\n");

        printf("The purpose of a loopback address is to allow a computer to communicate
        with itself without using a physical network connection.\n");

    } else {
```

```
        printf("\nClass : E\n");

        printf("IP Range : 240.0.0.0 - 255.255.255.255\n\n");


        printf("Purpose : Experimental\n\n");


        printf("Default Subnet Mask : Not Applicable\n");

        printf("A subnet mask is a 32-bit number used with an IP address to identify
        which part of the address represents the network and which part represents the
        host (device).\n\n");


        printf("Hosts per Network ID : Not Applicable\n");

        printf("Number of Networks : Not Applicable\n\n");


        printf("Loopback Address : NO\n");

        printf("A loopback address is a special IP address that a computer uses to
        communicate with itself for testing and troubleshooting network software
        without sending data over a physical network.\n");

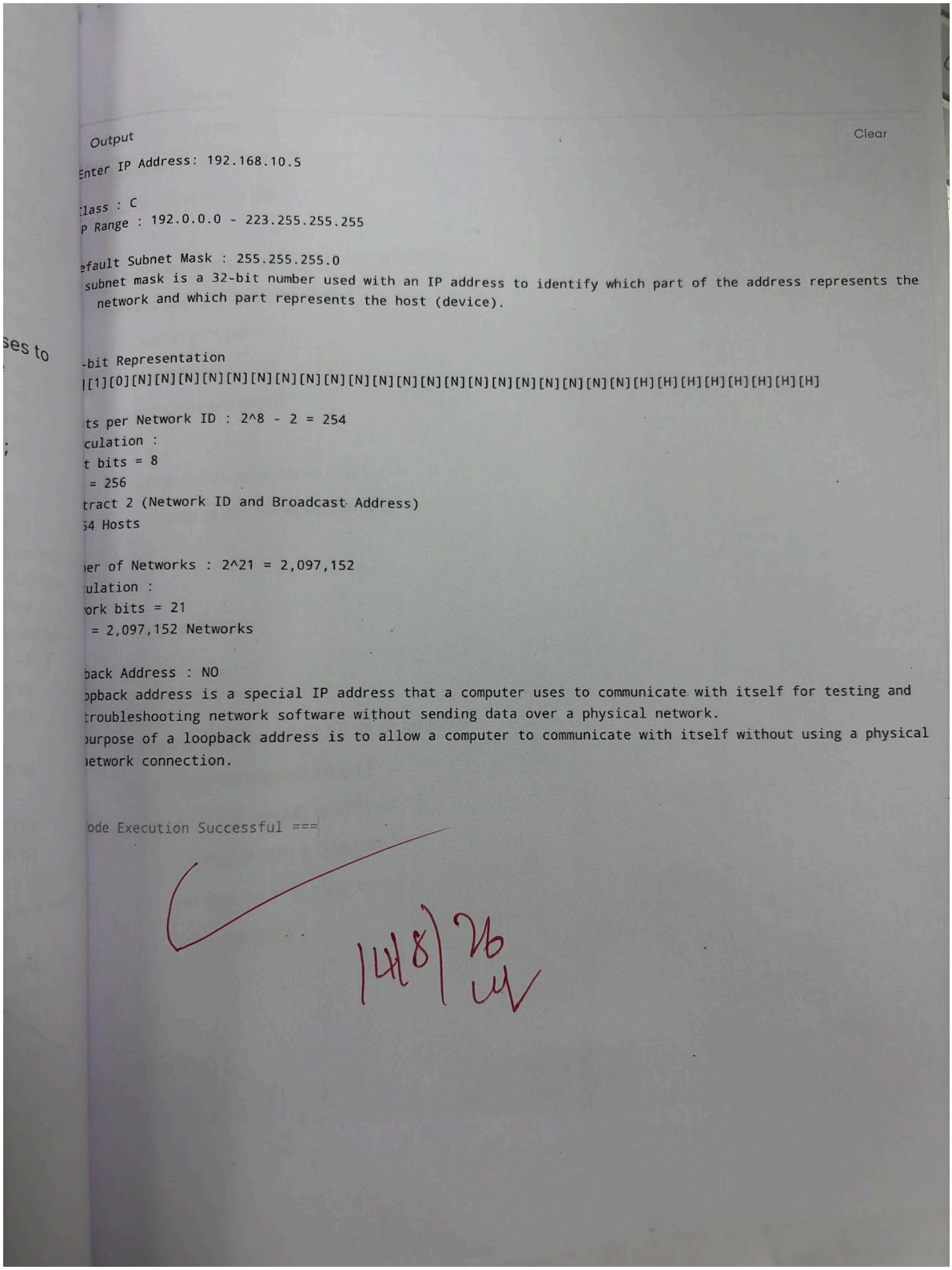
        printf("The purpose of a loopback address is to allow a computer to communicate
        with itself without using a physical network connection.\n");

    }


    return 0;

}
```

Output



Output Clear

Enter IP Address: 192.168.10.5

lass : C
P Range : 192.0.0.0 - 223.255.255.255

efault Subnet Mask : 255.255.255.0
subnet mask is a 32-bit number used with an IP address to identify which part of the address represents the network and which part represents the host (device).

-bit Representation
][1][0][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][N][H][H][H][H][H][H][H][H]

ts per Network ID : 2^8 - 2 = 254
culation :
t bits = 8
= 256
tract 2 (Network ID and Broadcast Address)
54 Hosts

er of Networks : 2^21 = 2,097,152
ulation :
ork bits = 21
= 2,097,152 Networks

back Address : NO
opback address is a special IP address that a computer uses to communicate with itself for testing and troubleshooting network software without sending data over a physical network.
urpose of a loopback address is to allow a computer to communicate with itself without using a physical network connection.

ode Execution Successful ===

14/8/26

2. Link: https://github.com/Kur00o/CNL/blob/main/ASS3.2.cpp

```cpp
#include <iostream>

#include <sstream>

#include <vector>

using namespace std;


vector<int> parseIP(string ip) {

   vector<int> parts;

   string temp;

   stringstream ss(ip);


   while (getline(ss, temp, '.')) {

       parts.push_back(stoi(temp));

   }


   return parts;

}


void printIP(unsigned int ip) {

   cout << ((ip >> 24) & 255) << "."

        << ((ip >> 16) & 255) << "."

        << ((ip >> 8) & 255) << "."
```

```
        << (ip & 255);
}

int main() {
   string ip;
   int requiredSubnets;

   cout << "Enter IP address: ";
   cin >> ip;

   cout << "Enter number of required subnets: ";
   cin >> requiredSubnets;

   vector<int> addr = parseIP(ip);

   char ipClass;
   int defaultBits;

   if (addr[0] >= 1 && addr[0] <= 126) {
       ipClass = 'A';
       defaultBits = 8;
   }
   else if (addr[0] >= 128 && addr[0] <= 191) {
       ipClass = 'B';
       defaultBits = 16;
   }
   else if (addr[0] >= 192 && addr[0] <= 223) {
       ipClass = 'C';
```

```
    defaultBits = 24;
}
else {
    cout << "Class D/E address. Not supported.\n";
    return 0;
}

cout << "\nIP Class: " << ipClass << endl;

int borrowedBits = 0;
int possibleSubnets = 1;

while (possibleSubnets < requiredSubnets) {
    borrowedBits++;
    possibleSubnets *= 2;
}

cout << "Borrowed bits: " << borrowedBits << endl;
cout << "Actual number of subnets: "
     << possibleSubnets << endl;

int totalNetworkBits = defaultBits + borrowedBits;
int hostBits = 32 - totalNetworkBits;

int addressesPerSubnet = 1 << hostBits;

cout << "Subnet prefix: /"
     << totalNetworkBits << endl;
```

```
cout << "Addresses per subnet: "
     << addressesPerSubnet << endl;

cout << "Usable IP addresses per subnet: "
     << addressesPerSubnet - 2 << endl;

unsigned int mask;

if (totalNetworkBits == 32)
    mask = 0xFFFFFFFF;
else
    mask = 0xFFFFFFFF << (32 - totalNetworkBits);

cout << "Subnet Mask: ";
printIP(mask);
cout << endl;

unsigned int ipValue =
    ((unsigned int)addr[0] << 24) |
    ((unsigned int)addr[1] << 16) |
    ((unsigned int)addr[2] << 8) |
    (unsigned int)addr[3];

unsigned int network = ipValue & mask;

cout << "\n-----------------------------------------------\n";
```

```
for (int i = 0; i < possibleSubnets; i++) {
    unsigned int subnetNetwork =
        network + (i * addressesPerSubnet);

    unsigned int broadcast =
        subnetNetwork + addressesPerSubnet - 1;

    unsigned int firstIP = subnetNetwork + 1;
    unsigned int lastIP = broadcast - 1;

    cout << "\nSubnet " << i + 1 << endl;

    cout << "Network Address : ";
    printIP(subnetNetwork);

    cout << "\nFirst IP        : ";
    printIP(firstIP);

    cout << "\nLast IP         : ";
    printIP(lastIP);

    cout << "\nBroadcast       : ";
    printIP(broadcast);

    cout << "\n-----------------------------------------------\n";
}

return 0;
```

```
}
```

Output

Enter IP address: 195.1.1.0

Enter number of required subnets: 10

IP Class: C

Borrowed bits: 4

Actual number of subnets: 16

Subnet prefix: /28

Addresses per subnet: 16

Usable IP addresses per subnet: 14

Subnet Mask: 255.255.255.240

Subnet 1

Network Address : 195.1.1.0

First IP : 195.1.1.1

Last IP : 195.1.1.14

Broadcast : 195.1.1.15

Subnet 2

Network Address : 195.1.1.16

First IP : 195.1.1.17

Last IP : 195.1.1.30

Broadcast : 195.1.1.31

Subnet 3

Network Address : 195.1.1.32

First IP : 195.1.1.33

Last IP : 195.1.1.46

Broadcast : 195.1.1.47

Subnet 4

Network Address : 195.1.1.48

First IP : 195.1.1.49

Last IP : 195.1.1.62

Broadcast : 195.1.1.63

Subnet 5

Network Address : 195.1.1.64

First IP : 195.1.1.65

Last IP : 195.1.1.78

Broadcast : 195.1.1.79

Subnet 6

Network Address : 195.1.1.80

First IP : 195.1.1.81

Last IP : 195.1.1.94

Broadcast : 195.1.1.95

Subnet 7

Network Address : 195.1.1.96

First IP : 195.1.1.97

Last IP : 195.1.1.110

Broadcast : 195.1.1.111

Subnet 8

Network Address : 195.1.1.112

First IP : 195.1.1.113

Last IP : 195.1.1.126

Broadcast : 195.1.1.127

Subnet 9

Network Address : 195.1.1.128

First IP : 195.1.1.129

Last IP : 195.1.1.142

Broadcast : 195.1.1.143

Subnet 10

Network Address : 195.1.1.144

First IP : 195.1.1.145

Last IP : 195.1.1.158

Broadcast : 195.1.1.159

Subnet 11

Network Address : 195.1.1.160

First IP : 195.1.1.161

Last IP : 195.1.1.174

Broadcast : 195.1.1.175

Subnet 12

Network Address : 195.1.1.176

First IP : 195.1.1.177

Last IP : 195.1.1.190

Broadcast : 195.1.1.191

Subnet 13

Network Address : 195.1.1.192

First IP : 195.1.1.193

Last IP : 195.1.1.206

Broadcast : 195.1.1.207

Subnet 14

Network Address : 195.1.1.208

First IP : 195.1.1.209

Last IP : 195.1.1.222

Broadcast : 195.1.1.223

Subnet 15

Network Address : 195.1.1.224

First IP : 195.1.1.225

Last IP : 195.1.1.238

Broadcast : 195.1.1.239

Subnet 16

Network Address : 195.1.1.240

First IP : 195.1.1.241

Last IP : 195.1.1.254

Broadcast : 195.1.1.255
